

```

    faceapi.matchDimensions(canvas,
displaySize)
    setInterval(async() => {
        const detections = await
faceapi.detectAllFaces(video, new
faceapi.TinyFaceDetectorOptions()).
withFaceLandmarks().
withFaceExpressions()
        const resizedDetections =
faceapi.resizeResults(detections,
displaySize)
        canvas.getContext('2d').
clearRect(0, 0, canvas.width, canvas.
height)
        faceapi.draw.
drawDetections(canvas,
resizedDetections)
        faceapi.draw.
drawFaceLandmarks(canvas,
resizedDetections)
        faceapi.draw.
drawFaceExpressions(canvas,
resizedDetections)
    }, 100)
})

```

In the above code, we load the pre-trained models using face-api. We are using four models: tinyFaceDetector, faceLandmark68Net, faceRecognitionNet and faceExpressionNet. Once these models are loaded, we call the `startVideo()` function, which starts capturing video from the webcam. The `navigator.getUserMedia` API is used to capture a video stream which we then assign to `video.srcObject`. Then we add an event listener to the video tag, which is fired when the video gets played. Using face-api, we get the video, detect the face, facial landmark and expression. We draw a rectangle around the detected face and facial landmark to denote eyebrows, eyes, nose and lips, and display the facial expressions of the user with text, under the drawn rectangle, with a detection accuracy score.

Testing the app

Now that we have added the code, let us start the server and test the final outcome. Run the following command

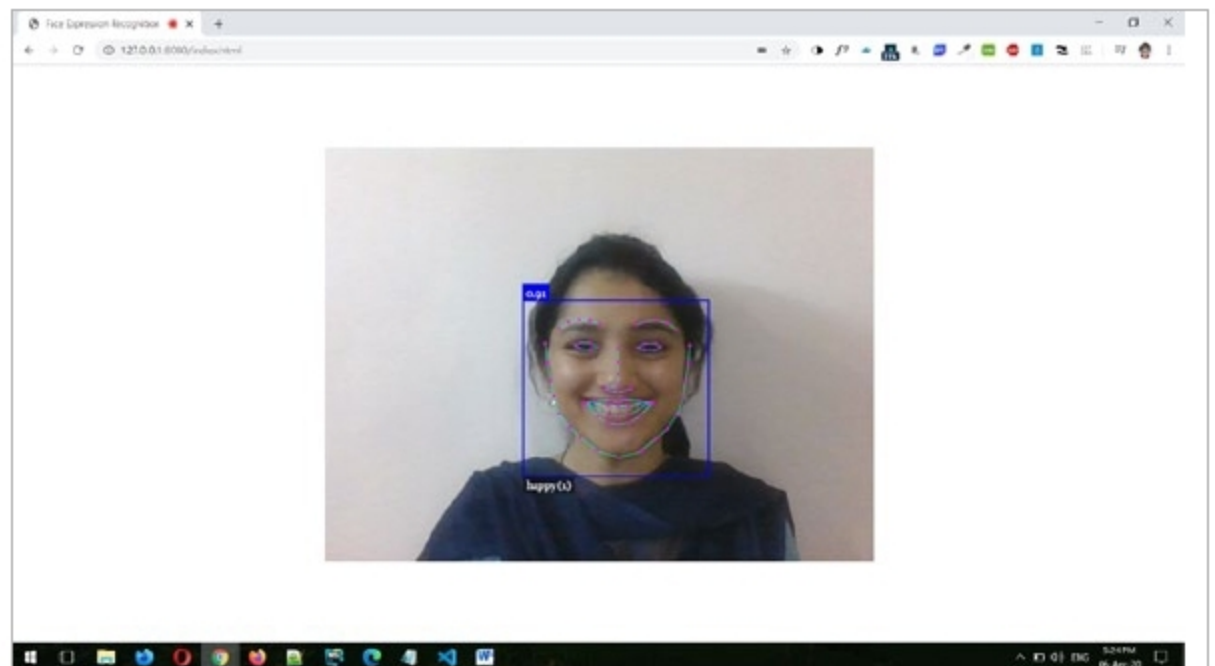


Figure 3: Face and expression getting detected and recognised

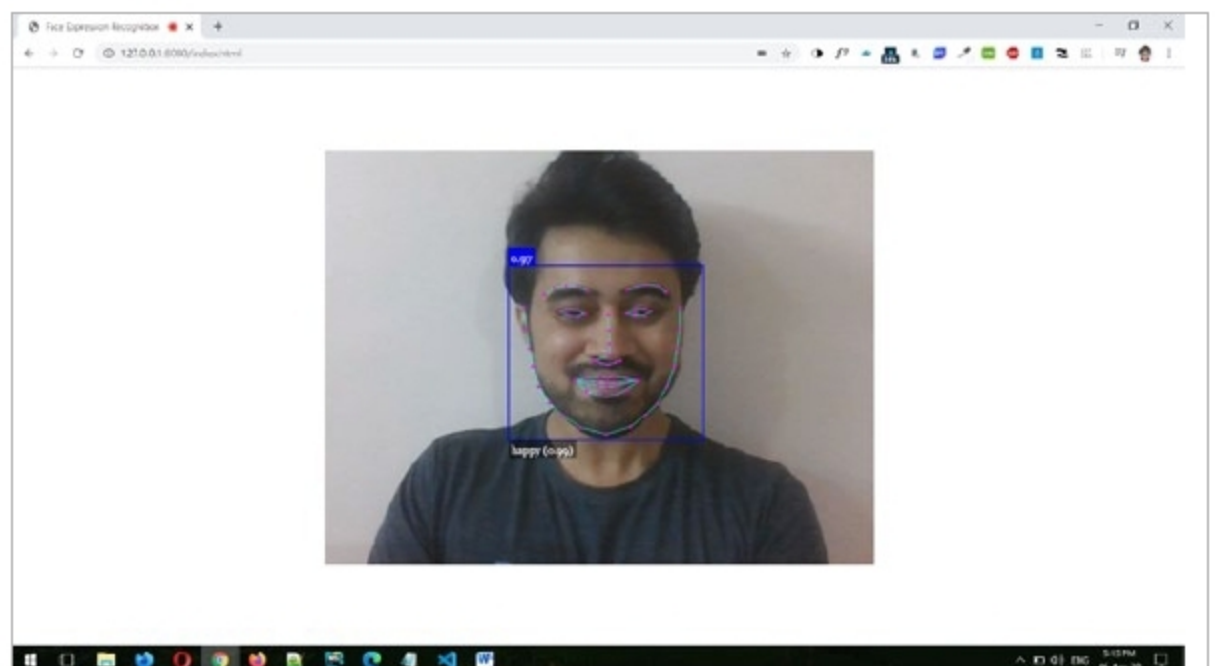


Figure 4: Face and happy expression getting detected and recognised

in the same directory:

```
http-server ./
```

This command will start the HTTP server. Now, load `http://127.0.0.1:8080/index.html` in your browser to run the project. If you see the output screen as shown in Figure 3, then you have successfully created a facial expression recognition app. You can now try detecting different facial expressions like happy, sad or angry.

As you can see in Figure 4, my face shows a happy expression. It detected and recognised my face and the expression I made, and displayed it with a rectangle and text in the webcam video.

I have uploaded this sample project to my GitHub repository at <https://github.com/aniketkudale/face-expression-recognition>. You may download it for reference. END 🐧

References

- [1] <https://www.tensorflow.org/>
- [2] <https://www.tensorflow.org/js>
- [3] <https://www.npmjs.com/package/face-api.js>

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